Geographic Information System

🟩 SLIP 1 – Practical 3

Dataset: `Practical_02\A\Data`
Raster file:
`gl_gpwv3_pdens_90ascii_one\glds90ag60.asc`

Aim

To add raster layers, set properties and symbology, perform raster styling and analysis, and apply raster mosaicking and clipping using QGIS.

Answer

Q. Add raster layers, set properties and symbology, apply raster styling, analysis, mosaicking and clipping

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A. Adding Raster Layer

1. Open **QGIS Desktop**.
2. From the **Menu Bar**, select
   Layer → Add Layer → Add Raster Layer.
3. Browse to the folder:
   `Practical_02\A\Data\gl_gpwv3_pdens_90ascii_one`
4. Select the raster file `glds90ag60.asc` and click **Add**.
5. The raster layer will be displayed on the map canvas.

B. Set Project CRS

6. Go to **Project → Properties**.
7. Select **CRS** from the left panel.
8. Set CRS as **WGS 84 (EPSG:4326)**.
9. Click **OK**.

C. Raster Properties and Symbology

10. In the **Layers Panel**, right-click on **glds90ag60.asc**.
11. Click **Properties**.
12. Select the **Symbology** tab.
13. Set:
    - Render type: **Singleband pseudocolor**
    - Color ramp: Suitable color ramp (e.g., Red to Green)
14. Click **Classify**.
15. Click **Apply** and then **OK**.

D. Raster Styling and Analysis

16. To perform raster analysis, go to **Raster → Raster Calculator**.
17. Enter a suitable raster expression (for example population threshold or difference).
18. Select output file location and name.
19. Click **OK**.
20. The analyzed raster output will be added to the map canvas.

E. Raster Mosaicking

21. Go to **Raster → Miscellaneous → Merge**.

22. Add multiple raster layers (if available).
23. Select output file location.
24. Click **Run**.
25. A mosaicked raster layer will be generated.

F. Raster Clipping

26. Go to **Raster → Extraction → Clip Raster by Extent**.
27. Select **Input Layer** as `glds90ag60.asc`.
28. Choose **Use Map Canvas Extent**.
29. Specify output file name and location.
30. Click **Run**.
31. The clipped raster layer will be displayed.

Result

The raster layer was successfully added, styled, analyzed, mosaicked, and clipped using QGIS.

🟩 SLIP 2

Practical: 4, 5, 6
Software: QGIS

Q.1 Add `ne_10m_admin_0_countries.zip` layer, save as shapefile, add raster `R17.tif` and show output with grid EPSG:27700

Steps

1. Open **QGIS Desktop**.
2. Go to **Layer → Add Layer → Add Vector Layer**.
3. Browse and select `ne_10m_admin_0_countries.zip`.
4. Click **Add** to load the vector layer.
5. Right-click on the layer → **Export → Save Features As**.
6. Select format **ESRI Shapefile**.
7. Choose file location and name.
8. Click **OK** to save the shapefile.
9. Go to **Layer → Add Layer → Add Raster Layer**.
10. Browse and select `R17.tif`.
11. Click **Add**.
12. Go to **Project → Properties → CRS**.
13. Set CRS as **EPSG:27700 (British National Grid)**.
14. Click **OK**.
15. Go to **View → Decorations → Grid**.
16. Enable grid and set suitable grid spacing.
17. Click **Apply → OK**.

Q.2 Open OpenStreetMap, search "Statue of Unity" and geo-reference the aerial image of Statue of Unity region

Steps

1. In the **Browser Panel**, expand **XYZ Tiles**.
2. Double-click **OpenStreetMap** to load it.
3. Use the **Search / Locator bar** and type **"Statue of Unity"**.
4. Zoom to the searched location.

5. Go to **Raster → Georeferencer**.
6. In Georeferencer window, click **File → Open Raster**.
7. Load the aerial image of **Statue of Unity region**.
8. Go to **Settings → Transformation Settings**.
9. Set:
    - Transformation type: **Thin Plate Spline**
    - Resampling method: **Nearest Neighbour**
    - Target CRS: Project CRS
10. Select output raster name and location.
11. Click **OK**.
12. Add Ground Control Points (GCPs) by matching locations with OSM.
13. Click **Start Georeferencing**.
14. The geo-referenced image will be added to the map canvas.

Q.3 Demonstrate the projection of WMS map

Steps

1. Go to **Layer → Add Layer → Add WMS/WMTS Layer**.
2. Click **New** and enter WMS connection details.
3. Click **Connect**.
4. Select a WMS layer and click **Add**.
5. Go to **Project → Properties → CRS**.
6. Change the CRS to another projection.
7. Observe that the WMS layer automatically reprojects.
8. Thus, projection of WMS map is demonstrated.

🟩 SLIP 3

Practical: 4 and 5
Software: QGIS

(Dataset as per Practical-4 and Practical-5 manuals)

Q.1 Open OpenStreetMap, search "University of Mumbai Fort Campus" and geo-reference aerial images of University of Mumbai Fort Campus and Churchgate

Steps

1. Open **QGIS Desktop**.
2. In the **Browser Panel**, expand **XYZ Tiles**.
3. Double-click **OpenStreetMap** to add it to the map canvas.
4. In the **Locator/Search bar**, type **"University of Mumbai Fort Campus"**.
5. Press **Enter** and zoom to the location.

Geo-referencing Aerial Image of University of Mumbai Fort Campus

6. Go to **Raster → Georeferencer**.
7. In the Georeferencer window, click **File → Open Raster**.
8. Select the aerial image of **University of Mumbai Fort Campus**.
9. Go to **Settings → Transformation Settings**.
10. Set:
    - Transformation type: **Thin Plate Spline**
    - Resampling method: **Nearest Neighbour**
    - Target CRS: Project CRS

11. Select output raster file name and location.
12. Click **OK**.
13. Add Ground Control Points (GCPs) by matching features with OpenStreetMap.
14. Click **Start Georeferencing**.
15. The geo-referenced image will appear on the map.

Geo-referencing Aerial Image of Churchgate

16. Again open **Raster → Georeferencer**.
17. Click **File → Open Raster**.
18. Load the aerial image of **Churchgate**.
19. Set transformation settings same as above.
20. Add Ground Control Points using OpenStreetMap reference.
21. Click **Start Georeferencing**.
22. The geo-referenced Churchgate image will be added to the map canvas.

Q.2 Import elevation raster `10n060e_20101117_gmted_mea300.tif`, extract hilly area of North India by extent and perform terrain analysis

Steps

1. Go to **Layer → Add Layer → Add Raster Layer**.
2. Browse and select `10n060e_20101117_gmted_mea300.tif`.
3. Click **Add** to load the elevation raster.
4. The elevation map of India will be displayed.
5. Zoom to **North India hilly region** using map navigation tools.

6. Go to **Raster → Extraction → Clip Raster by Extent**.
7. Select **Input Layer** as `10n060e_20101117_gmted_mea300.tif`.
8. Choose **Use Map Canvas Extent**.
9. Specify output raster name and location.
10. Click **Run** to extract the required area.

Terrain Analysis

11. Go to **Raster → Analysis → DEM (Terrain Models)**.
12. Select the clipped raster as input.
13. Choose terrain analysis type (Slope / Aspect / Relief).
14. Specify output file location.
15. Click **Run**.
16. The terrain analysis output for hilly area of North India will be displayed.

Result

OpenStreetMap was used to geo-reference aerial images, and elevation raster data was clipped and analyzed to study terrain characteristics of hilly areas in North India.

🟩 SLIP 4

Practical: 4 and 5
Software: QGIS

(As per Practical-4 and Practical-5 manuals)

Q.1 Open OpenStreetMap, search "University of Mumbai Fort Campus" and geo-reference the aerial image of

University of Mumbai Fort Campus

Steps

1. Open **QGIS Desktop**.
2. In the **Browser Panel**, expand **XYZ Tiles**.
3. Double-click **OpenStreetMap** to add it to the map canvas.
4. In the **Search / Locator bar**, type **"University of Mumbai Fort Campus"**.
5. Press **Enter** and zoom to the searched location.
6. Go to **Raster → Georeferencer**.
7. In the Georeferencer window, click **File → Open Raster**.
8. Select the aerial image of **University of Mumbai Fort Campus**.
9. Go to **Settings → Transformation Settings**.
10. Set:
    - Transformation type: **Thin Plate Spline**
    - Resampling method: **Nearest Neighbour**
    - Target CRS: Project CRS
11. Select output raster file name and location.
12. Click **OK**.
13. Add Ground Control Points (GCPs) by matching identifiable locations with OpenStreetMap.
14. Click **Start Georeferencing**.
15. The geo-referenced image will be added to the map canvas.

Q.2 Import elevation raster

`10n060e_20101117_gmted_mea300.tif`, extract

hilly area of North India by extent and perform Hill Shade surface analysis

Steps

1. Go to **Layer → Add Layer → Add Raster Layer**.
2. Browse and select `10n060e_20101117_gmted_mea300.tif`.
3. Click **Add** to load the elevation raster.
4. The surface elevation map of India will be displayed.
5. Zoom to the **hilly region of North India**.
6. Go to **Raster → Extraction → Clip Raster by Extent**.
7. Select **Input Layer** as `10n060e_20101117_gmted_mea300.tif`.
8. Choose **Use Map Canvas Extent**.
9. Specify output raster file name and location.
10. Click **Run** to extract the required raster area.

Hill Shade Analysis

11. Go to **Raster → Analysis → DEM (Terrain Models)**.
12. Select the **clipped raster** as input.
13. Choose **Hillshade** as the analysis type.
14. Set output file name and location.
15. Click **Run**.
16. The hillshade surface representing the hilly terrain of North India will be displayed.

Result

The aerial image of University of Mumbai Fort Campus was successfully geo-referenced using OpenStreetMap, and hillshade analysis was performed on elevation data of North India.

🟩 SLIP 5

Practical: 2
Software: QGIS

(As per Practical-2: Creating and Managing Vector Data

Final QGIS Practical 2 File

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Q.1 Create three vector layers to store point locations of college, restaurants and hospitals. Add roads and railway tracks. Change colour and size of point markers to differentiate restaurants

Steps

1. Open **QGIS Desktop** and start a new project.
2. Go to **Layer → Create Layer → New Shapefile Layer**.
3. Select **Geometry Type** as **Point**.
4. Add attribute fields such as **Name** and **Type**.
5. Set CRS as **WGS 84 (EPSG:4326)**.
6. Save the layer as **College.shp**.
7. Repeat steps 2 to 6 to create point layers **Restaurants.shp** and **Hospitals.shp**.
8. Go to **Layer → Create Layer → New Shapefile Layer**.
9. Select **Geometry Type** as **Line**.
10. Add attributes and save the layer as **Roads.shp**.
11. Repeat to create **Railway_Tracks.shp**.

12. Enable **Toggle Editing** and add point features for college, restaurants and hospitals.
13. Add line features for roads and railway tracks.
14. Right-click on **Restaurants** layer → **Properties → Symbology**.
15. Change the **colour and size** of restaurant point symbols.
16. Click **Apply → OK**.

Q.2 Create a map using the above layers and add labels, legend, scale and suitable title

Steps

1. Go to **Project → New Print Layout**.
2. Enter a suitable layout name and click **OK**.
3. In the Print Layout window, select **Add Item → Add Map**.
4. Draw the map area on the layout canvas.
5. Go to **Add Item → Add Label**.
6. Enter a suitable **map title** and format font size and style.
7. Go to **Add Item → Add Legend** and place it on the layout.
8. Uncheck **Auto Update** and adjust legend entries if required.
9. Go to **Add Item → Add Scale Bar** and place it appropriately.
10. Adjust scale bar units and style.
11. Save the layout or export it as image/PDF.

Result

Vector layers were created, symbology was applied, and a complete map with title, legend and scale bar was prepared.

🟩 SLIP 6

Practical: 3 and 5
Software: QGIS

Datasets used:

- `Practical_02\IndiaAdminBoundry\IND_adm0.shp`
- `Practical_04\Data\GMTED2010N10E060_300`
- `10n060e_20101117_gmted_mea300.tif`

(As per Practical-3 and Practical-5 manuals)

Q.1 Open `IND_adm0.shp` and geo-reference the Indian political map at appropriate location

Steps

1. Open **QGIS Desktop**.
2. Go to **Layer → Add Layer → Add Vector Layer**.
3. Browse and select `IND_adm0.shp` .
4. Click **Add** to load the India boundary layer.
5. Zoom to the India region on the map canvas.
6. Go to **Raster → Georeferencer**.
7. In the Georeferencer window, click **File → Open Raster**.
8. Select the scanned **Indian Political Map** image.
9. Go to **Settings → Transformation Settings**.
10. Set:
    - Transformation type: **Thin Plate Spline**
    - Resampling method: **Nearest Neighbour**

- Target CRS: **WGS 84 (EPSG:4326)**

11. Select output raster file name and location.
12. Click **OK**.
13. Add Ground Control Points (GCPs) by matching political boundaries with `IND_adm0.shp`.
14. Click **Start Georeferencing**.
15. The geo-referenced political map will be displayed correctly over India.

Q.2 Import elevation raster `10n060e_20101117_gmted_mea300.tif`, extract hilly area of North India by extent and perform Terrain analysis

Steps

1. Go to **Layer → Add Layer → Add Raster Layer**.
2. Browse and select `10n060e_20101117_gmted_mea300.tif`.
3. Click **Add** to load the elevation raster of India.
4. Zoom to the **hilly region of North India** (Himalayan region).
5. Go to **Raster → Extraction → Clip Raster by Extent**.
6. Select input layer as `10n060e_20101117_gmted_mea300.tif`.
7. Choose **Use Map Canvas Extent**.
8. Specify output raster file name and location.
9. Click **Run**.

Terrain Analysis

10. Go to **Raster → Analysis → DEM (Terrain Models)**.
11. Select the clipped raster as input.

12. Choose terrain analysis type (**Slope / Aspect / Relief**).
13. Select output location and file name.
14. Click **Run**.
15. Terrain analysis output for North India will be displayed.

Q.3 Import elevation raster `10n060e_20101117_gmted_mea300.tif`, extract hilly area of North India and perform Hill Shade analysis

Steps

1. Ensure the raster layer `10n060e_20101117_gmted_mea300.tif` is loaded.
2. Zoom again to the **hilly area of North India**.
3. Go to **Raster → Extraction → Clip Raster by Extent**.
4. Select **Use Map Canvas Extent**.
5. Save the clipped raster file.
6. Go to **Raster → Analysis → DEM (Terrain Models)**.
7. Select the clipped raster as input.
8. Choose **Hillshade** as analysis type.
9. Set output raster file name and location.
10. Click **Run**.
11. Hillshade surface representing hilly terrain of North India will be generated.

Result

The Indian political map was geo-referenced successfully, and terrain as well as hillshade analysis was performed on elevation data of North India.

🟩 SLIP 7

Practical: 2
Dataset: No dataset
Software: QGIS

Q.1 Create a map of your own residential area

Steps

1. Open **QGIS Desktop**.
2. Go to **Project → New**.
3. Set the Project CRS to **WGS 84 (EPSG:4326)**.
4. Go to **Layer → Create Layer → New Shapefile Layer**.
5. Create the following layers:
   - **Polygon layer** for Gardens
   - **Line layer** for Roads
   - **Line layer** for Highways
   - **Line layer** for Railway Tracks
   - **Point layer** for Restaurants
6. Add appropriate attribute fields while creating each layer.
7. Enable **Toggle Editing** and digitize features representing the residential area.
8. Save all layers and stop editing.

Q.2 Perform the following queries on the above layers

(a) Add a new column `Garden_area` and calculate area of each garden

1. Right-click on **Garden layer** → **Open Attribute Table**.
2. Click **Toggle Editing**.
3. Click **Add Field**.
4. Set:
   - Field name: `Garden_area`
   - Type: **Decimal Number**
5. Click **OK**.
6. Open **Field Calculator**.
7. Select **Update existing field** → `Garden_area` .
8. Enter expression:

   $area

9. Click **OK** and save edits.

(b) Show the largest garden

1. Open **Garden Attribute Table**.
2. Click **Select Features Using Expression**.
3. Enter:

   "Garden_area" = maximum("Garden_area")

4. Click **Select Features**.
5. The largest garden will be highlighted on the map.

(c) Show all railway tracks that intersect with highways

1. Go to **Vector → Research Tools → Select by Location**.
2. Set:
   - Select features from: **Railway Tracks**
   - Predicate: **Intersects**

- Comparing to: **Highways**

3. Click **Run**.
4. Intersecting railway tracks will be selected.

(d) Find all restaurants inside a garden

1. Go to **Vector → Research Tools → Select by Location**.
2. Set:
   - Select features from: **Restaurants**
   - Predicate: **Within**
   - Comparing to: **Gardens**
3. Click **Run**.
4. Restaurants inside gardens will be selected.

(e) Find all railway tracks that do not intersect with highways

1. After selecting intersecting railway tracks, go to **Edit → Invert Selection**.
2. The selected features now represent railway tracks **not intersecting highways**.

Result

A residential area map was created and required spatial queries were successfully performed using QGIS.

🟩 SLIP 8

Practical: 2
Dataset: No dataset
Software: QGIS

Q.1 Create a map of your own residential area

Steps

1. Open **QGIS Desktop**.
2. Go to **Project → New**.
3. Set the Project CRS to **WGS 84 (EPSG:4326)**.
4. Go to **Layer → Create Layer → New Shapefile Layer**.
5. Create required layers:
    - **Polygon layer** for Residential Area / Gardens
    - **Line layer** for Roads
    - **Line layer** for Railway Tracks
    - **Point layer** for Hospitals
6. Add suitable attribute fields for each layer.
7. Enable **Toggle Editing**.
8. Digitize features representing the residential area.
9. Save all layers and stop editing.

Q.2 Create a map using the above layers and add required map elements

Steps

1. Go to **Project → New Print Layout**.
2. Enter a layout name and click **OK**.
3. In Print Layout window, select **Add Item → Add Map**.
4. Draw the map area on the layout canvas.

Add North Pointer

5. Go to **Add Item → Add Picture**.
6. Select a suitable **North Arrow** image.
7. Place it on the map.

Add Legend and Scale

8. Go to **Add Item → Add Legend**.
9. Place legend at appropriate location.
10. Go to **Add Item → Add Scale Bar**.
11. Place scale bar and set units.

Add Symbols

12. Right-click on **Hospitals layer → Properties → Symbology**.
13. Select **any suitable symbol** to represent hospitals.
14. For Railway Stations, select one feature.
15. Change its symbol to **Metro Station symbol**.

Save Map as JPEG

16. Go to **Layout → Export as Image**.
17. Select format **JPEG**.
18. Set resolution to **1500 dpi**.
19. Choose file location and click **Save**.

Result

A residential area map was created with north pointer, legend, scale, custom symbols, and exported in JPEG format with high resolution.

🟩 SLIP 9

Practical: 4
Software: QGIS

Q.1 Create a map based on given assumptions

Steps

1. Open **QGIS Desktop**.
2. Go to **Project → New**.
3. Set Project CRS as **WGS 84 (EPSG:4326)**.
4. Create required layers using
   Layer → Create Layer → New Shapefile Layer:
   - **Polygon layer** for Farms
   - **Point layer** for Wells
   - **Line layer** for Paths
   - **Line layer** for Highways
5. Add suitable attribute fields for each layer.
6. Enable **Toggle Editing**.
7. Digitize features based on given assumptions.
8. Save all layers.

Q.2 Perform the following queries on above layers

(a) Show all farms that contain wells

1. Go to **Vector → Research Tools → Select by Location**.

2. Set:
    - Select features from: **Farms**
    - Predicate: **Contains**
    - Comparing to: **Wells**
3. Click **Run**.
4. Farms containing wells will be selected.

(b) Calculate area of each farm and show the largest farm

1. Open **Farm layer Attribute Table**.
2. Click **Toggle Editing**.
3. Click **Add Field**.
4. Set:
    - Field name: `Farm_area`
    - Type: **Decimal Number**
5. Click **OK**.
6. Open **Field Calculator**.
7. Update existing field **Farm_area**.
8. Enter expression:

   $area

9. Click **OK** and save edits.
10. Click **Select Features Using Expression**.
11. Enter:

    "Farm_area" = maximum("Farm_area")

12. Click **Select Features**.
13. The largest farm will be highlighted.

(c) Show all paths that intersect with highways

1. Go to **Vector → Research Tools → Select by Location**.
2. Set:
   - Select features from: **Paths**
   - Predicate: **Intersects**
   - Comparing to: **Highways**
3. Click **Run**.
4. Intersecting paths will be selected.

Result

The map was created based on assumptions and spatial queries were successfully performed.

🟩 SLIP 10

Practical: 2
Software: QGIS

Q.1 Create vector data (minimum 5 layers: line, polygon, point) and prepare appropriate database

Steps

1. Open **QGIS Desktop**.
2. Go to **Project → New**.
3. Set Project CRS to **WGS 84 (EPSG:4326)**.
4. Go to **Layer → Create Layer → New Shapefile Layer**.
5. Create the following layers:

- **Polygon layer** (e.g., Residential Area)
- **Polygon layer** (e.g., Gardens)
- **Line layer** (e.g., Roads)
- **Line layer** (e.g., Railway Tracks)
- **Point layer** (e.g., Hospitals / Restaurants)

6. While creating each layer, add appropriate attribute fields such as:
    - Name
    - Type
    - ID
7. Click **OK** to create each layer.
8. Enable **Toggle Editing** for each layer.
9. Digitize required features on the map canvas.
10. Save edits and stop editing for all layers.

Q.2 Demonstrate the use of analytical tools: Distance Matrix and Points in Layer

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(a) Distance Matrix

1. Go to **Vector → Analysis Tools → Distance Matrix**.
2. Select **Input point layer** (e.g., Hospitals).
3. Select **Target point layer** (e.g., Restaurants).
4. Choose measurement type as **Linear distance**.
5. Specify output table file name and location.
6. Click **Run**.
7. Distance matrix table will be generated showing distances between points.

(b) Points in Layer

1. Go to **Vector → Analysis Tools → Count Points in Polygon**.
2. Select **Polygon layer** (e.g., Gardens).
3. Select **Point layer** (e.g., Restaurants).
4. Choose output file name and location.
5. Click **Run**.
6. A new polygon layer will be created with a field showing number of points inside each polygon.

Result

Vector layers were created successfully, database attributes were prepared, and analytical tools Distance Matrix and Points in Polygon were demonstrated in QGIS.